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SET 4 — Product Inventory Management

According to your experiment sheet, Set 4 uses **Product Inventory** data with Product ID, Product Name, and Quantity Available. The four tasks are a bar chart, stacked bar chart, scatter plot, and Tableau dashboard.

Dataset

Product	Quantity
Product A	250
Product B	175
Product C	300
Product D	200
Product E	220

AIM

To analyze product inventory data using **R programming** and visualize the quantity available for different products using charts.

ALGORITHM

1. Start RStudio.
2. Create the product inventory dataset.
3. Store Product Name, Quantity Available, Product Category, and Price.
4. Create a **bar chart** to visualize the quantity of each product.
5. Create a **stacked bar chart** to show inventory quantity by product category.
6. Create a **scatter plot** between product price and quantity available.
7. Add suitable titles and axis labels.
8. Calculate the correlation between price and quantity.
9. Display the graphs and result.
10. Stop.

R CODE

```
inventory <- data.frame(  
  Product=c("Product A","Product B","Product C","Product D","Product E"),  
  Quantity=c(250,175,300,200,220),  
  Category=c("Electronics","Clothing","Electronics","Clothing","Electronics"),  
  Price=c(20,15,18,25,22)
```

)

1. Bar Chart

```
barplot(inventory$Quantity,  
        names.arg=inventory$Product,  
        main="Product Inventory",  
        xlab="Product",  
        ylab="Quantity Available")
```

2. Stacked Bar Chart

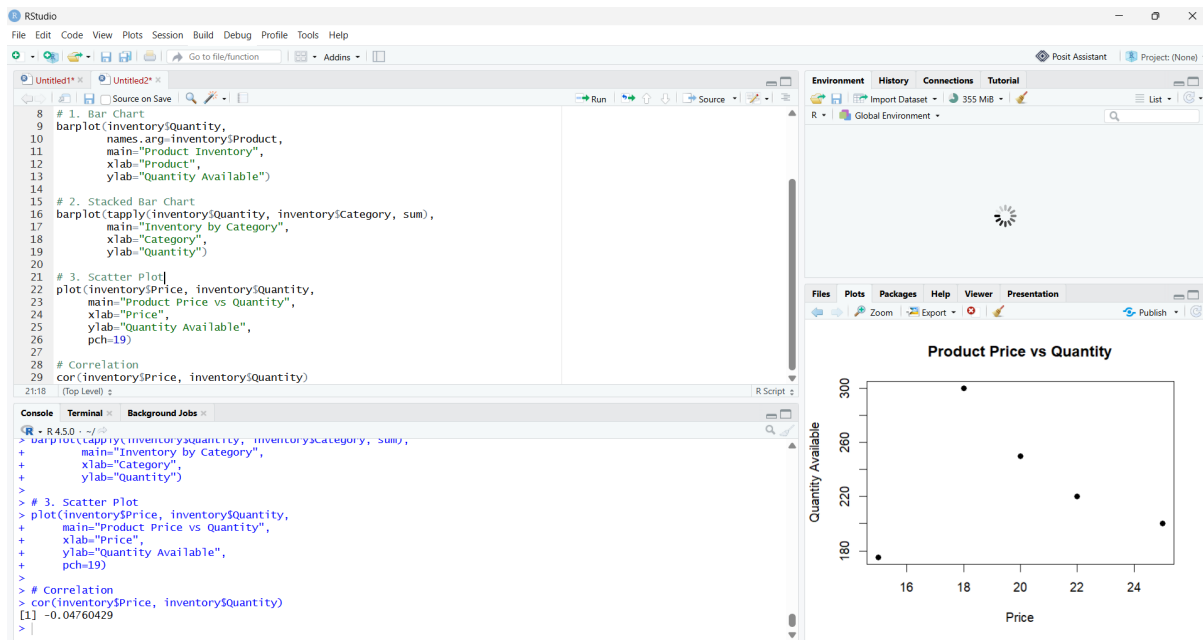
```
barplot(tapply(inventory$Quantity,inventory$Category,sum),  
        main="Inventory by Category",  
        xlab="Category",  
        ylab="Quantity")
```

3. Scatter Plot

```
plot(inventory$Price,inventory$Quantity,  
     main="Product Price vs Quantity",  
     xlab="Price",  
     ylab="Quantity Available",  
     pch=19)
```

Correlation

```
cor(inventory$Price,inventory$Quantity)
```



RESULT

Thus, the **Product Inventory Management** data was successfully analyzed using R programming. The **bar chart**, **stacked bar chart**, and **scatter plot** were successfully created to visualize product quantities and the relationship between product price and quantity available.